

Rachel M. Garcia

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EDUCATION

Simpson College

Bachelor of Arts in Mathematics

Indianola, IA

Expected May 2027

- **Relevant Coursework:** Real Analysis, Adv Data Modeling, Intro to Epidemiology, Basic Infectious Disease
- **Research Interests:** Infectious disease epidemiology, spatial health disparities, and social determinants of health in Latin American populations

RESEARCH EXPERIENCE

Researcher, SIAM-Simons Program

Inter American University

June 2026 – July 2026

Bayamón, Puerto Rico

- [To be updated with project details]

Research Assistant, SYLFF Research Grant

Simpson College

May 2025 – July 2025

Indianola, IA

- Collaborated on a manuscript with an Assistant Professor of Economics, adapting dissertation-level research for academic publication
- Performed comprehensive analysis on a 1000 respondent survey dataset in STATA, including variable construction, regression modeling, and diagnostic testing to examine how student debt shapes labor market decisions
- Produced publication-ready figures and regression tables demonstrating statistically significant associations between debt burden and occupational risk preferences

Researcher, Greta A. & Albert H. Bryan Program

Simpson College

June 2024 – August 2024

Indianola, IA

- Conducted geospatial analysis of Down syndrome birth rates across all 32 Mexican states, identifying statistically meaningful regional clusters and health disparities using R spatial analysis packages (sf, leaflet, ggplot2)
- Quantified associations between socioeconomic factors and Down syndrome birth outcomes, contributing epidemiological evidence to an interdisciplinary Pediatrics and Economics research team
- Designed interactive and dynamic visualizations in R that enabled non-technical individuals to explore spatial correlations between health outcomes and social determinants

CAMPUS LEADERSHIP & ACTIVITIES

President

Latinos Unidos

December 2023 – May 2026

- Led student organization of **100+** members promoting Latino culture, organizing events, and community engagement in the Des Moines Metropolitan Area

Undergraduate Assistant

Speech and Debate

August 2025 – May 2026

- Assisted in coaching **50+** students on researching diverse topics for advocacy and debate purposes

Undergraduate Assistant

Mathematics Department

August 2024 – Present

- Worked alongside professors in the Mathematics Department to recruit and support current and future mathematics majors

TECHNICAL SKILLS

Statistical Software: R(ggplot2, dplyr, leaflet, sf), STATA(regression analysis, data cleaning), Python(pandas, matplotlib, statsmodels, scipy)

Methods: Spatial Analysis, Multivariate regression, Hypothesis Testing, Data Modeling

Tools: LaTeX, Git, AWS, Microsoft Excel, Tableau

Languages: English (native), Spanish (elementary)